MASKINHALLEN · USER MANUAL · VERSION 1.4

BETONG

POLYRHYTHM TECHNO DRUM MACHINE · FREE

Chops sounds into skewed, polyrhythmic techno that drags and stumbles but always lands on the one. Free, full version.



VST3 · AU · LV2 · STANDALONE
MACOS · WINDOWS · LINUX

MASKINHALLEN

CONTENTS

01 Welcome
02 Install
03 Your first beat in one minute
04 The window
05 The top bar
06 Sounds, pads and chopping
07 The techno filter
08 The sequencer
09 The MOTOR: beats that live
10 Effects and performance
11 MIDI and automation
12 Recipes
13 FRIKTION: micro timing and interlock
14 New in 1.2 – 1.4
15 Troubleshooting

01 WELCOME

BETONG is a drum machine and sampler for hard, hypnotic techno. You load sounds onto 16 pads, play them from an 8-track step sequencer, and push everything through a filter, a dub delay and a rumble that ducks under the kick.

The idea is simple: give every track its own length and its own rate. A 16-step hat against a 13-step perc against a 5-tuplet clap. They drift apart and meet again, so the groove keeps moving without you touching anything.

- 16 pads and 215 original 24-bit sounds, built around a core of clean, fat drum-machine drums, free to use in your own tracks
- 8-track sequencer with accents and ghost notes, pitch per step, ratchets, probability and swing per track
- Four scenes, a fill button, and **SVAJ** and **GLITCH** to make the machine sway and break up
- Euclidean magic: patterns that rotate and breathe by themselves, accent, ratchet and pitch layers, tracks that fill each other's gaps, and a ring view
- A techno filter on every track with drive, envelope and bit crush
- Dub delay, rumble, master sweep, clipper and stutter
- 38 kits, 169 patterns and 54 effect presets in genre folders, plus dice that invent new ones

ABOUT THE SWEDISH LABELS

BETONG was built in Stockholm, so the controls have Swedish names. This manual writes every label exactly as it looks on screen, like `SPELA`, with the English meaning next to it the first time. Here are the words you will see most:

ON SCREEN	ENGLISH	ON SCREEN	ENGLISH
`SPELA` / `STOPP`	Play / Stop	`MÖNSTER`	Pattern
`SPÅR`	Track	`ALLT`	Everything
`GALENSKAP`	Madness	`Ångra`	Undo
`Lås`	Lock	`Bibliotek`	Library
`Hacka till pads`	Chop to pads	`Baklänges`	Reverse
`SÄND`	Send	`NIVÅ`	Level
`VOLYM`	Volume	`PANORERING`	Pan
`SVEP`	Sweep	`GRUS`	Grit
`UT`	Output	`STORLEK`	Size
`SVAJ`	Sway	`DENSITET`	Density
`Scen`	Scene	`Kopiera till...`	Copy to...
`Rotera själv`	Rotate by itself	`Andas`	Breathe
`Fyll luckorna i`	Fill the gaps in	`Stark / Svag`	Strong / Weak (ghost)

THE WINDOW



02 INSTALL

Unzip the file for your system. You get a folder called **BETONG**. Then run the installer for your system and restart your DAW (or rescan plug-ins). BETONG shows up as an instrument.

MACOS

1. Double-click `BETONG-macOS.zip` in Downloads.
2. Open Terminal and run:
   `bash ~/Downloads/BETONG/installera.command`
3. Quit your DAW completely and start it again.

VST3, AU and a standalone app. macOS 11 or newer, Intel or Apple Silicon. The installer copies the plug-ins and removes macOS's block on downloaded plug-ins.

WINDOWS

1. Right-click `BETONG-Windows.zip` and choose **Extract all**.
2. Open the `BETONG` folder and right-click `installera.bat` → **Run as administrator**.
3. If Windows says "Windows protected your PC": **More info** → **Run anyway**.

VST3 (in C:\Program Files\Common Files\VST3) and a standalone .exe. 64-bit Windows 10 or 11.

LINUX

1. Unzip `BETONG-Linux.zip`.
2. Run `bash ~/Downloads/BETONG/installera-linux.sh`. It copies the plug-ins to `~/.vst3` and `~/.lv2`.
3. Rescan plug-ins in your DAW (Bitwig, Reaper, Ardour and others).

VST3 and LV2, 64-bit x86 Linux (built on Ubuntu 22.04).

> SWEDISH OR ENGLISH
>
> The machines have Swedish names on the panel. One click on `SV | EN` in the header switches labels, menus and tooltips to English. Your choice is remembered for every MASKINHALLEN machine.

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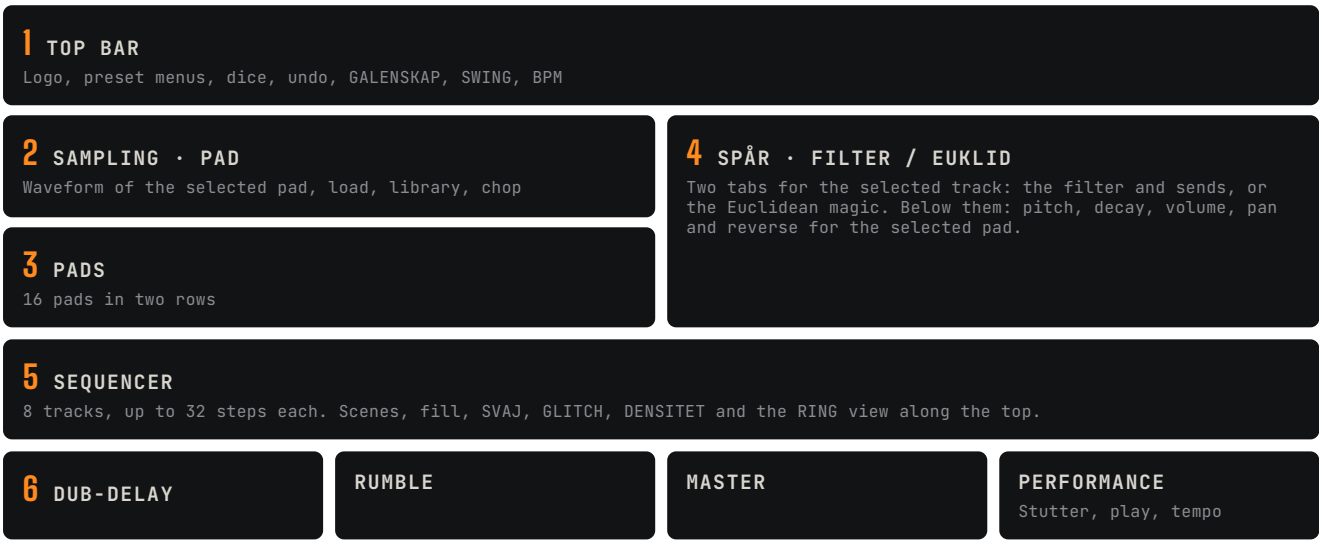
03 YOUR FIRST BEAT IN ONE MINUTE

1. Put BETONG on an instrument track in your DAW and open it.
2. Press play in the DAW. BETONG follows the DAW tempo and starts playing right away. In the standalone app, press `SPELA` *(play)* at the bottom right.
3. Pick a kit in the first menu at the top, and a pattern in the second. The menus are sorted in genre folders. Start with the kits under **Rena grundkit** *(clean basic kits)*.
4. Press the `ALLT` *(everything)* dice. You get a new kit, pattern and effects in one go.
5. Turn `GALENSKAP` *(madness)* up and press `ALLT` again. It gets weirder. Didn't like it? Press `Ångra` *(undo)*.
6. Hold `STUTTER` on a build-up, and turn `SVEP` *(sweep)* left to filter the whole mix.

> GOOD TO KNOW
>
> Everything in BETONG is saved with your DAW project: sounds, pads, all four scenes, the Euclidean settings and every knob. The window can be resized from the bottom right corner.

04 THE WINDOW

BETONG has six areas, from top to bottom.



Selected pad and selected track. Many controls act on what is selected. Click a pad to select it. Click the name at the left of a sequencer row to select that track. Selecting a track also selects the pad it plays, and clicking a pad selects the track that plays it.

About MASKINHALLEN. Click the BETONG logo at the top left to read where it all started: Docklands, a rattling boatyard in Stockholm.

05 THE TOP BAR

PRESET MENUS

The three menus load ready-made settings. They only replace their own part, so you can mix a kit from one style with a pattern from another.

MENU	WHAT IT LOADS	
Kit (first menu)	16 sounds onto the pads.	38 kits
Pattern (second)	A full sequencer pattern, all 8 tracks, plus swing. Goes into the scene you are on.	169 patterns

MENU	WHAT IT LOADS	
Effects (third)	Delay, rumble, clipper, master grit, and the sends and distortion on each track.	54 presets

Each menu opens in genre folders: Techno, Peak time, Hardgroove, Tech house, Tribal, Minimal, Dub, Polyrytm *(polyrhythm)*, Euklid-magi *(Euclidean magic)* and the classic sets.

THE DICE

BUTTON	WHAT IT DOES
`KIT`	New sounds on all pads that are not locked. Kicks stay on pads 1–2, hats on 3–4, claps on 5–6, and so on, so the result still makes sense.
`FX`	Nudges the delay, rumble and filters in new directions.
`MÖNSTER`	*(pattern)* A new crooked pattern on all tracks that are not locked, with new lengths, rates and swing.
`ALLT`	*(everything)* All three at once.
`Ångra`	*(undo)* Goes back one dice roll. Remembers the last 20.

`GALENSKAP` *(madness)* sets how wild the dice are. Low: straight four-on-the-floor, normal lengths, sounds in their usual place. High: odd lengths like 5, 7 and 13, tuplets, ratchets, probability, pitched and reversed sounds, and strange sounds on unexpected pads.

`Lås` *(lock)* on a sequencer row protects that track and its pad from the dice. Lock your kick and bass, and roll the rest.

SWING AND TEMPO DISPLAY

`SWING` pushes every second step later. It works on 1/16, 1/8, 1/16T and 1/32 tracks. The 5- and 7-tuplet tracks are never swung, which makes them sound even more off against the rest.

At the top right you see the tempo. It says **BPM · VÄRD** *(host)* when the DAW is playing, and BPM when BETONG uses its own tempo.

06 SOUNDS, PADS AND CHOPPING

LOADING A SOUND

There are five ways to get a sound onto a pad:

- **Right-click a pad** and pick a built-in sound from the categories, or choose `Ladda fil...` *(load file)*.
- **Drag a file** from Finder or Explorer onto a pad, or onto the waveform to load it to the selected pad.
- **Drag several files** at once onto a pad. They fill that pad and the ones after it, in order.
- Press `Ladda ljud` *(load sound)* above the waveform to pick a file for the selected pad.

- Press `Bibliotek` *(library)* and click a sound in the list.

File types: WAV, AIFF, FLAC and OGG. MP3 also works on Mac.

THE LIBRARY

The library opens over the waveform and pads. Pick a category on the left, then click a sound on the right. It loads straight onto the selected pad and plays, so you can click through sounds while the beat runs. Press `Stäng` *(close)* when you are done.

The built-in categories are kicks, hats and cymbals, clap and snare, perc, bass, stabs and chords, leads, textures and drones, voices, FX, and loops at 130 BPM. All tonal sounds are in F minor, so they fit together.

Your own sounds. Press `+ Lägg till mapp` *(add folder)* and pick a folder with samples. Its sounds show up under **Egna mappar** *(my folders)*. BETONG remembers your folders in every project. `Ta bort mapp` *(remove folder)* removes the chosen folder from the list. It does not delete any files.

THE PADS

Click a pad to play it and select it. Each pad shows its sound name, its number, and which track plays it, for example **PAD 5 · SPÅR 3**. The coloured bar on top matches the track colour. Pads flash when they play.

- **Hi-hat choke:** pad 3 (closed hat) cuts off pad 4 (open hat), just like on a real drum machine.
- **One voice per track:** a new hit on a track cuts off the last one. This keeps basses and acid lines tight.

SHAPING A SOUND

The waveform shows the sound on the selected pad. The bright part is what the pad plays.

- **Drag the orange edges** to move the start and end.
- **Click inside the waveform** to hear the pad.
- Thin lines with numbers show other pads that use the same sound, for example after chopping.

Under the filter on the right there are five controls for the selected pad: `PITCH` (±24 semitones), `DECAY` (shortens the sound), `PAD VOLYM` (−30 to +6 dB), `PAD PAN` and `Baklänges` *(reverse)*. Use pad volume to balance a kit.

CHOPPING A SOUND ONTO THE PADS

Chopping cuts one sound into pieces and puts them on pads 1–16. It is the fastest way to turn a loop or a vocal into a new beat.

1. Load a loop or a long sound onto any pad and select it.
2. Choose how to cut in the menu above the waveform.
3. Press `Hacka till pads` *(chop to pads)*.

MODE	HOW IT CUTS
`Anslag` *(hits)*	Finds the hits in the sound and cuts at each one. The **känslighet** *(sensitivity)* bar sets how many hits it finds. Drag it right for more slices.
`Jämnt 8`	8 equal slices on pads 1–8.
`Jämnt 16`	16 equal slices on pads 1–16.

Chopping replaces the sounds on the pads it fills. You can also right-click a pad and choose `Hacka detta ljud till alla pads` *(chop this sound to all pads)*, which uses the hits mode.

07 THE TECHNO FILTER

Every one of the 8 tracks has its own filter, distortion and sends. The panel on the right shows the selected track, for example **SPÅR 3**. Make sure the `FILTER` tab is chosen. The `EUKLID` tab is explained in chapter 08.

FILTER TYPE

`LP 24` and `LP 12` are low-pass (dark), `HP` is high-pass (thin), `BP` is band-pass (nasal) and `Notch` cuts a narrow band. `Av` *(off)* bypasses the filter.

KNOBS

KNOB	WHAT IT DOES	RANGE
`CUTOFF`	Where the filter cuts.	30 Hz–18 kHz
`RESONANS`	Boost at the cutoff. Turn it high and the filter starts to scream and distort.	0–100 %
`DRIVE`	Distortion into the filter.	0–100 %
`ENVELOPE`	Opens the filter on every hit, then closes it again. Accented steps open it more.	0–100 %
`DECAY`	How fast the envelope closes.	5–1500 ms
`CRUSH`	Bit crusher, down to 4 bit.	0–100 %
`SÄND DELAY`	How much of the track goes to the dub delay.	0–100 %
`SÄND RUMBLE`	How much goes to the rumble. Track 1 sends 90 % from the start.	0–100 %
`VOLYM`	Track volume. Above 100 % is boost.	0–150 %
`PANORERING`	Left and right.	L–R
`SWING SPÅR`	*(track swing)* Swing for this track only. **Global** follows the main SWING. Shuffle the shaker while the kick stays straight.	Global, 50–88 %

> TRY THIS
> Hats on LP 24, CUTOFF low, ENVELOPE high, DECAY short. Every hat gets a little wah. Then turn RESONANS up slowly.

08 THE SEQUENCER

Eight tracks, one per row. Each track plays one pad, with its own length and rate. This is where the polyrhythms come from.

ONE ROW, LEFT TO RIGHT

CONTROL	WHAT IT DOES
Number and name	Click to select the track. Shows the sound it plays.
Pad	Which of the 16 pads the track plays.
Length	How many steps before the track starts over, 1–32. Drag up or down, or use the arrows.
Rate	How fast the steps go: **1/16**, **1/8**, **1/16T** (triplets), **1/32**, **5-del** (5 per beat) and **7-del** (7 per beat).
`E`	Euclidean mode. See below.
Hits, rotation	Only used in Euclidean mode.
Steps	The step grid. Dim boxes are past the track's length.
`M` `S` `Lås`	Mute, solo, and lock against the dice.

EDITING STEPS

DO THIS	TO GET	SHOWS AS
Click	Step on or off. Drag to paint many steps.	`filled`
`Shift` + click	Strength. Each click steps normal → strong (louder, opens the filter more) → weak (a soft ghost note) → normal.	`white bar / dim`
Scroll over a step	Pitch of that step in semitones. Use it to play basslines and melodies on bass and stab pads. You hear each new note.	`+3`
`Alt` + click	Probability. Each click steps 100 → 75 → 50 → 25 %.	`75`
`Cmd` + click (Mac) `Ctrl` + click (Win)	Ratchet: the step repeats 2, 3 or 4 times. Keep clicking to step through.	`×3`

DO THIS	TO GET	SHOWS AS
Right-click	Menu with all of the above: `Styrka` *(strength)*, `Ton` *(pitch, −12 to +12)*, probability, ratchet, plus `Töm spåret` *(clear track)* and `Fyll varje steg` *(fill every step)*.	

Clicking a step past the end of a track makes the track longer, up to that step.

POLYRHYTHMS IN PRACTICE

Set the kick to 16 steps. Set a perc to 13 steps. The perc lands one step earlier against the kick each bar, and only lines up again after 13 bars. Now set a clap to **5-del**: five steps per beat against the kick's four. Small changes in length and rate give grooves that keep shifting for minutes.

EUCLIDEAN MODE

Press `E` on a row. Now you choose only **how many hits** and BETONG spreads them as evenly as possible over the track's length. **Rotation** shifts the whole pattern forward. Classic shapes: 3 hits in 8 steps, 5 in 8, 7 in 16. Editing a step by hand turns Euclidean mode off again.

PLAYING

BETONG follows your DAW: it starts and stops with the DAW and uses its tempo. Without a DAW, or in the standalone app, press `SPELA`. It changes to `STOPP` while running. While the DAW plays, the button says `DAW` and is locked.

SCENES AND FILL

A B C D above the tracks are four scenes: four complete patterns in the same kit. Click a scene while the beat plays and it blinks until the next bar, then switches exactly on the downbeat. When stopped it switches at once. `Kopiera till...` *(copy to...)* copies the scene you see to another one, so you can make a variation and keep the original.

Hold `FILL` for a drum roll: the clap, snare and tom tracks (pads 5–8) roll in sixteenths and get louder towards the end of the bar, and the other tracks double their hits. The kick keeps going. Let go and the pattern is back.

SVAJ, GLITCH AND DENSITET

Three bars along the top of the sequencer change how the pattern plays, without changing the pattern itself. Turn them down and everything is clean again.

BAR	WHAT IT DOES
`SVAJ`	*(sway)* Hits drag a little behind the beat, their strength varies and the pitch drifts a few cents, like an old machine that wanders. The kick sways only half as much, so the groove holds.
`GLITCH`	Random bursts, reversed hits, pitch jumps, cut-off hits and dropped hits. The kick glitches rarely.
`DENSITET`	*(density)* Thins out or fills up the whole beat. Euclidean tracks add or remove hits and keep them evenly spread. Other tracks drop or add soft hits. Turn it up into a drop, down into a break.

EUCLIDEAN MAGIC: THE EUKLID TAB

Select a track and choose the `EUKLID` tab in the panel on the right. Everything here is calculated while the beat plays. The pattern itself never changes.

CONTROL	WHAT IT DOES
`ROTERA SJÄLV`	*(rotate by itself)* The track moves one step every 1, 2, 3, 4, 5, 8 or 16 bars. A Euclidean pattern always stays Euclidean, so it keeps grooving while it slowly turns.
`ANDAS`	*(breathe)* The number of hits sweeps up and down by the amount you set, over 4–32 bars. Build-ups and breaks happen by themselves. On normal tracks it makes the density breathe instead.
`ACCENTLAGER`	*(accent layer)* Its own Euclidean pattern of strong hits, with its own length. Give it a length that differs from the track, for example 3 in 8 over a 16-step track, and the accents walk around.
`RATCHETLAGER`	*(ratchet layer)* Its own pattern of bursts: how many, over how many steps, and 2–4 hits per burst.
`TONLAGER`	*(pitch layer)* Its own pattern of pitch jumps, in semitones. On a bass track the line tangles itself up and untangles again, because the layer lands in a new place each time round.
`FYLL LUCKORNA I`	*(fill the gaps in)* The track only plays where another track is silent. If the track has no steps of its own, it fills every gap. Gives tight, interlocking African patterns that never collide.
`Nollställ magin`	*(reset the magic)* Turns all of the above off for the track.

THE RING VIEW

Press `RING` above the tracks. The tracks turn into rings inside each other, with track 1 on the outside. Each dot is a step: big dots are hits, ringed dots are strong, faint dots are weak. A white circle shows where each track is playing, so you can watch a 7-step ring turn against a 16-step ring.

- **Drag around a ring** to rotate that track.
- **Drag straight up or down, or scroll,** to add or remove hits. This turns on Euclidean mode.
- **Double-click a ring** to turn Euclidean mode on or off.

Next to the rings each track shows its length, rate and any Euclidean magic it has. Press `RING` again to go back to the step grid. The pattern folder **Euklid-magi** has 14 examples to start from.

09 THE MOTOR: BEATS THAT LIVE

Press `MOTOR` above the tracks and the beat starts to live on its own. Tracks come and go with the energy, the pattern changes a little every phrase, filters breathe and sounds get thrown into the delay. Your pattern never changes: the MOTOR

calculates a living copy while it plays. The same seed always sounds the same, so DAW loops and exports sound exactly like what you heard.

THE MOTOR TAB

Open the `MOTOR` tab in the panel on the right. At the top: `MOTOR PÅ` (on), the form, `LÄGE` (ready-made motor settings), the seed and `NY` (a new seed).

FORM	WHAT HAPPENS
Bara liv	*(just life)* No song form. Everything plays, with small changes and breathing filters.
Fri	*(free)* Endless waves of energy, with a break and a lift now and then.
Kort 64, Låt 128, Afterhour 256	*(short, song, afterhour)* A full track over 64, 128 or 256 bars: intro, build, peak, break, lift, peak 2 and outro. Bar 1 in your DAW is the start of the intro.

The **energy curve** shows the whole form with the play position. Click it to switch to the **evolution map**: one row per track and one column per bar. Brighter means more hits, dark means the track rests, and a cold line on top means the MOTOR has changed the pattern in that bar.

KNOB	WHAT IT DOES
`RÖRELSE`	*(movement)* How often things change: from every 32 bars to every 2 bars.
`DJUP`	*(depth)* How big each change is. One hit is added, removed, moved, made stronger or weaker, or retuned. After four changes the original pattern comes back, and sometimes an old motif returns, slightly changed.
`KAOS`	*(chaos)* Pulls everything at once: more movement and depth, a rougher energy curve, more drops and some glitch.
`FILTER`	Filters breathe slowly, close before every change and open when the new part comes in. Resonance peaks when the energy is high, a darker master in the break and a rising high-pass in the lift.
`DUB`	At the end of a phrase one hit is thrown into the dub delay and fades out.
`PUMP`	Sidechain from the kick and drive that grow with the energy.
`SVÄNG`	*(swing)* Velocity waves with their own lengths, micro-timing (hats drag, the kick stays on the grid) and swing that drifts slowly.

In the sequencer you see the MOTOR work live: a cold frame is an added hit, a crossed-out step is a removed one, a cold dot is a changed velocity or pitch, and a resting track turns dark. Tracks with `Lås` (lock) are left alone, and empty tracks stay empty.

TRIG CONDITIONS

Right-click a step and choose **Villkor** (condition). **1:4** plays only the first time round out of four, **3:4** only the third, and so on up to 8:8. **FILL** plays only while `FILL` is held, **!FILL** only when it is not. **PRE** plays if

the previous condition on the track was true, **!PRE** if it was not. The condition is shown at the top of the step.

GAME OF LIFE AND CALL AND RESPONSE

Right-click the name of a track.

- **Game of Life:** the steps become cells that are born and die once per bar. *Puls* (rule 90) ripples, *Kaos* (rule 30) is wild, *Liv* is a gentle one-dimensional Life. The pattern you drew is the first generation, and every 16 bars life starts over from it. New hits are soft ghost notes.
- **Fråga och svar** *(call and response)*: set one track to *Fråga* (call) and another to *Svar* (response). They take turns in blocks of 1, 2, 4 or 8 bars.

BREED AND MORPH

- `AVLA` *(breed)* pairs the scene that plays with another scene. Each track inherits its base from one parent, the steps are crossed over at a random point and a small mutation is added. The child goes into a third scene and starts on the next bar. `Ångra` takes it back.
- `MORF` slides the steps over to another scene, step by step. Right-click it to choose the scene. Automate it in your DAW for long transitions.

DRAG OUT AS MIDI

Drag the `MIDI` button on the MOTOR tab into your DAW. What plays, with the MOTOR, life, conditions, morph, swing and micro-timing, becomes a MIDI file with one track per sequencer track, starting at bar 1. Drum tracks use pad notes (C1 = pad 1). Bass, stabs and leads use real pitches with F2 as the root. Click the button to choose the length or to save the file.

AI HELP WITHOUT KEYS

1. Press `AI` and choose **Kopiera AI-fråga** (copy AI question). BETONG puts a ready-made question on the clipboard. It describes the format, the sounds on your pads and the pattern that plays.
2. Paste it into any AI chat (Gemini, ChatGPT, Claude) and write what you want at the top: a style, a feeling, a track or an artist.
3. Copy the whole answer and choose **Klistra in AI-svar** (paste AI answer). The pattern, swing and MOTOR settings load. Locked tracks are left alone, and `Ångra` takes it back.

No account, no key and nothing is sent anywhere by BETONG. The AI only knows what has been written about a track, it cannot listen to it, so treat the answer as a starting point.

PATTERNS TO START FROM

The pattern folder **Motor och liv** uses conditions, Game of Life and call and response. Load one, press `MOTOR` and choose a `LÄGE`, for example *Hypnotisk minimal* or *Plastik-drift*.

10 EFFECTS AND PERFORMANCE

The bottom row has two send effects, the master section and a performance panel.

DUB-DELAY

A ping-pong delay with a filter in the echoes. Feed it with `SÄND DELAY` on each track.

CONTROL	WHAT IT DOES
Time menu	1/8, 1/8 punkt *(dotted)*, 3/16, 1/8T, 5/16 or 1/4. Always in time with the beat.
`Freeze`	Holds what is in the delay and loops it forever. Great for breaks.
`ÅTERKOPPL.`	*(feedback)* How long the echoes last.
`TON`	*(tone)* Darker or thinner echoes.
`WOBBLE`	Tape wobble on the echoes.
`NIVÅ`	*(level)* How loud the delay is in the mix.

RUMBLE

The low, distorted reverb that sits under the kick in warehouse techno. Every hit on **track 1** ducks it, so the kick always punches through. Track 1 is the kick in every kit.

CONTROL	WHAT IT DOES
`STORLEK`	*(size)* Size of the reverb.
`DRIVE`	Distortion on the reverb.
`CUTOFF`	Low-pass, 40–300 Hz. Keeps only the low end.
`DUCK`	How much the kick pushes the rumble down.
`NIVÅ`	*(level)* How loud the rumble is.

MASTER

CONTROL	WHAT IT DOES
`SVEP`	*(sweep)* A DJ filter on everything. Left is low-pass, right is high-pass, centre is off.
`GRUS`	*(grit)* Crunch on the whole mix.
`CLIPPER`	Pushes the mix into a soft clipper for a louder, harder sound.
`UT`	*(output)* Output level, −24 to +6 dB.

The master also makes everything below about 120 Hz mono, so the low end stays solid on big systems.

PERFORMANCE

`STUTTER` loops a short piece of the output while you hold it down. Pick the length in the menu next to it: 1/4, 1/8, 1/16 or 1/32. `TEMPO` sets BETONG's own tempo (60–200 BPM) when it plays without a DAW.

11 MIDI AND AUTOMATION

MIDI OUT

BETONG sends its sequencer out as MIDI while it plays: everything you hear, with the MOTOR, life, swing and micro-timing. Pick the mode in the `MIDI OUT` menu next to MIDI IN.

- **Track → channel 1–8:** track 1 on channel 1, track 2 on channel 2, and so on. Drum tracks use pad notes (C1 = pad 1). Bass, stabs and leads use real pitches with F2 as the root. Record each channel to its own track, or let a track drive another synth.
- **BETONG map, channel 1:** every hit as a pad note (C1 = pad 1), to drive a second BETONG or a drum rack.
- **Off:** no MIDI out.

In Bitwig, Reaper and Cubase you pick BETONG as the MIDI input on another track. Notes from your keyboard are not passed through, only the sequencer's hits. When the transport stops, all notes are released.

PLAY THE PADS FROM A KEYBOARD

BETONG listens to MIDI notes 36–51, so 16 keys play pads 1–16. Note 36 is called C1 in most DAWs (C3 in FL Studio). Velocity sets how loud the pad plays. You can play pads by MIDI while the sequencer runs, for fills on top of the pattern.

NOTES	PADS
36–43	Pads 1–8 (top row)
44–51	Pads 9–16 (bottom row)

AUTOMATION

Every knob and switch can be automated from your DAW: the effects, master, swing, `SVAJ`, `GLITCH`, `DENSITET`, `FILL`, all 8 track filters and track swing, and pitch, decay, volume, pan and reverse on all 16 pads. In the DAW they have Swedish names, for example *Spår 3 cutoff* (track 3 cutoff) or *Pad 5 pitch*.

12 RECIPES

HYPNOTIC LOOP IN FIVE MINUTES

1. Load the kit **Industriell 909** and turn `GALENSKAP` to about half.
2. Lock the kick track with `Lås`. Roll `MÖNSTER` until you hear a groove you like.
3. Set one perc track to 7 or 13 steps, and another to **5-del**.
4. Send that perc to the delay with `SÄND DELAY`, and set the delay to 3/16.
5. Automate `SVEP` slowly over 32 bars.

EUCLIDEAN HYPNOSIS

1. Load the kit **Betong Grund** and the pattern **Grund 130**.
2. Select the perc track, press `E` and set 5 hits in 13 steps.
3. On the `EUKLID` tab set `ROTERA SJÄLV` to every 2nd bar and `ANDAS` to 3 over 16 bars.
4. Select the bass track and give it a `TONLAGER` of 2 in 7 at +12.
5. Open `RING` and watch it turn. Ride `DENSITET` into the drop.

REBUILD A LOOP

1. Drag a drum loop onto pad 1.
2. Choose `Anslag` and press `Hacka till pads`.
3. Point the tracks at the slices and program a new order. Turn on `Baklänges` on one slice.

A WHOLE TRACK FROM ONE LOOP

1. Load the pattern **Livets Hats** and the kit **Plastik Minimal**.
2. Press `MOTOR`, choose the form **Låt 128** and the mode *Hypnotisk minimal*.
3. Press play in your DAW from bar 1 and let it run. Press `NY` for a new version of the same track.
4. When you like it, drag the `MIDI` button into your DAW, or record the audio.

THE BREAK

1. Turn up `SÄND DELAY` on the clap, then press `Freeze` on the delay.
2. Turn `SVEP` slowly to the right to thin out the whole mix.
3. Hold `STUTTER` at 1/16 for the last bar, then let go and turn `SVEP` back to the centre.

13 FRIKTION: MICRO TIMING AND INTERLOCK

The FRIKTION row above the effects stops BETONG from sitting on a stiff grid. The kick stays locked as the anchor, everything else is pulled back, leans on each other and stumbles in a controlled way. Everything on zero = the straight grid. Projects saved before FRIKTION open with it on zero.

CONTROL	WHAT IT DOES
POCKET	The base drag, 0–45 ms behind the grid, for everything except the kick. Hats drag fully, perc a bit more, clap a bit less, bass and tones only slightly. Each track also has its own factor, SLÄP SPÅR, in the track panel.
TRÖGHET	Inertia. Hits in a row drag more and more (+8, +18, +30 ms), as if pulled through syrup, and snap back at the next pause. Ratchets drag too.
HUMANIZE	Up to ±3 ms of analogue flutter per hit. It repeats every cycle, so it breathes without becoming noise.
INTERLOCK	The tracks listen to each other: clap/rim before perc before hats. Up to 50 %, a weaker hit that lands on the same step as a stronger one is ducked in level and decay. Above 50 %, it moves 1/32 or 1/64 later as a quieter, shorter, darker ghost note, so the drums answer each other in a rolling conversation.
MIKRO-RATCHET	Chance that a hit bounces 2–3 times, faster and quieter, like a scratched needle or a late quintuplet. Never on the beats.
VEL → DECAY	Ghost notes and late hits get shorter and darker; accents open their tails.
TAPE GLUE	The kick pushes down the tails of everything else (pump), and the whole mix goes through a warm, asymmetric tape saturation. The machine breathes on the off-beat.
VÄVEN	The weave. One row per track: the line is the grid, the dot is where the hit landed, the trail shows the last hits. S = ghost note, D = ducked, R = micro ratchet.
DICE // POCKET	New drag, inertia, interlock and tape, own drag factors per track and new polymetric lengths (11, 13, 15, 16). Always something that swings. Ångra takes it back.
POLY	Gives the drum tracks their own lengths by role: kick 16, hats 15, clap/rim 13, perc 11 and 13. They drift apart and meet again only after hundreds of bars. Locked tracks are left alone.

14 NEW IN 1.2 – 1.4

SLÄPP IN: A LOOP BECOMES A KIT, A PATTERN AND A GROOVE (1.4)

Drop in a drum loop and BETONG takes it apart for you.

1. Press `SLÄPP IN` in the SAMPLING area (or choose it in a pad's menu) and pick a loop, or drag a loop onto the button.
2. BETONG listens to the loop in three bands (bass, mid and treble), so hits that land at the same time are still found.
3. The hits are sorted into kick, clap, closed hat, open hat and two percussion sounds. The cleanest hit of each kind becomes a pad.
4. Where the hits were becomes the pattern on tracks 1–6, and the loop's drag and swing are set in FRIKTION.
5. The whole loop is also placed on pad 16 / track 8 (muted), so you can bring the original back in.

From there it is a normal BETONG beat: press the dice, start the MOTOR, swap single sounds. The groove of the original loop stays.

MIDI OUT (1.3)

Next to MIDI IN there is a `MIDI UT` selector. The sequencer's hits are sent out as notes, with all swing and micro timing, sample accurate:

- **Off**, **track → channel 1–8** (one MIDI channel per track), or **BETONG map, channel 1**.
- Drum tracks start at C1 = pad 1, tonal tracks use F2 as the root, the same notes as the MIDI drag-out.
- All notes are released when the transport stops. Incoming MIDI is not echoed.

MIDI IN (1.2)

Set `Sekvenser` to **MIDI IN** to switch off the internal sequencer and play the pads from your DAW, GLAPP or KOLV. The MIDI map can be BETONG, GM drums or GM channel 10 only. MIDI hits trigger the track's filter envelope, pump and tape glue just like the sequencer does.

SOUND QUALITY

One-shots fade out smoothly instead of clicking, slices that start mid-sound or play reversed get a 1 ms fade-in, and voice stealing takes the quietest voice.

15 TROUBLESHOOTING

BETONG does not show up	Quit the DAW completely and start it again, or rescan plug-ins. On Mac, run the installer again: it removes macOS's quarantine on the plug-in. On Windows, check that `BETONG.vst3` is in C:\Program Files\Common Files\VST3.
macOS says the plug-in is damaged	That is macOS's block on downloaded files. Run `bash ~/Downloads/BETONG/installera.command` again.
Windows protected your PC	Click **More info** → **Run anyway**. The plug-ins are not signed with a paid certificate.
The key does not work	Paste the whole key, including the dashes. A key for one machine only unlocks that machine; a bundle key unlocks all.
No sound	Press play in the DAW: the machines follow the transport. Instruments go on an instrument track, effects on an audio track or bus.

Anything else? Ask through the store, and the person who built the machine answers.

MASKINHALLEN